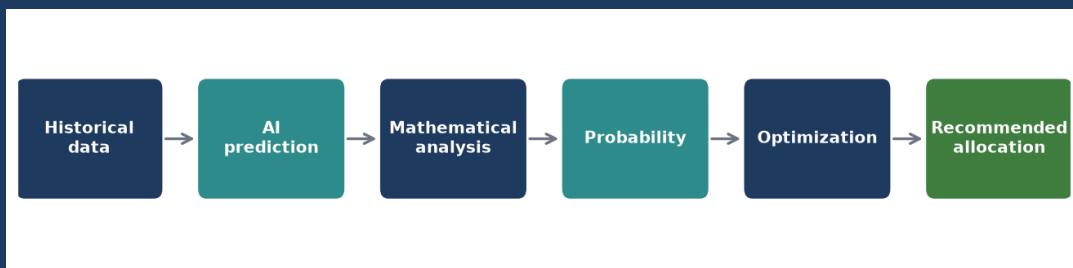


MetroMind AI

Intelligent Public Transport Demand Prediction & Resource Optimization System

AI-Driven Public Transport Demand Prediction and Mathematical Resource Optimization



Understand → Predict → Quantify uncertainty → Optimize → Recommend

1. The basic idea

MetroMind AI

Intelligent Public Transport Demand Prediction & Resource Optimization System

Imagine a city has buses running on several routes.

At 8:00 AM:

- Route A has 1,500 passengers waiting
- Route B has 400
- Route C has 900

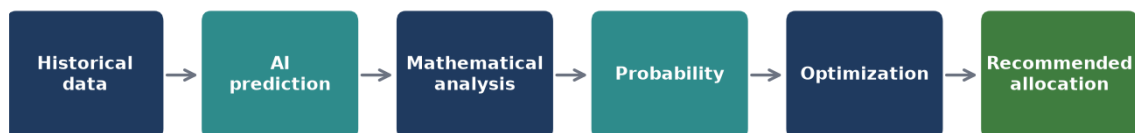
But the transport operator has a limited number of buses.

The real question is:

How many passengers are likely to travel on each route, and how should limited buses be allocated to reduce overcrowding and unnecessary empty capacity?

MetroMind AI tries to answer that question using:

Historical data → AI prediction → Mathematical analysis → Probability → Optimization → Recommended allocation



The heart of the project: historical data flows through AI and mathematics into a recommended allocation.

That is the heart of the project.

2. What real-world problem are you solving?

Public transport systems face several problems:

Overcrowding

Too many passengers during peak periods can lead to uncomfortable and inefficient service.

Underutilization

Some routes may have buses running with relatively low demand.

Demand uncertainty

Passenger demand changes depending on:

- time

- day
- weekday/weekend
- holidays
- weather
- location
- special events
- historical trends

Limited resources

You cannot simply add unlimited buses.

So the actual optimization problem is:

Given predicted passenger demand and limited transport resources, what is the best allocation of buses/resources?

This makes it a genuine optimization problem rather than just a prediction app.

3. Who would use the system?

You can define three roles.

Passenger

A passenger can:

- view route information
- see predicted crowd levels
- see expected demand
- identify less crowded travel times
- potentially receive alternative route/time suggestions

Transport operator

The operator is the main user. They can:

- view demand predictions
- see overcrowding risk
- see route statistics
- inspect trends
- simulate different resource allocations
- receive recommended bus allocation

Administrator

The administrator can:

- upload/update datasets
- configure routes
- manage stations
- manage vehicle capacities
- monitor system performance

4. What exactly happens inside MetroMind?

Suppose you upload historical data:

Date	Time	Route	Passengers
12-08-2026	08:00	R1	1420
12-08-2026	09:00	R1	1180
12-08-2026	08:00	R2	620
12-08-2026	09:00	R2	540
...	...	...	...

You can also have:

- Weather
- Day of week
- Holiday
- Rainfall
- Temperature
- Special event

The system processes this information.

Step 1 — Data analysis

The system asks:

What factors are related to passenger demand?

For example:

- Rainfall vs Passenger Demand
- Time vs Passenger Demand
- Weekday vs Passenger Demand
- Temperature vs Passenger Demand

This is where correlation becomes useful.

Your Maths syllabus explicitly includes Karl Pearson correlation, Spearman rank correlation, and regression.

5. AI prediction

Now the AI model learns patterns from historical data.

For example:

Input:

- Route = R1
- Time = 08:00
- Day = Monday
- Rainfall = High
- Holiday = No

Prediction:

Passenger demand = 1,560

The AI's job is:

Predict future passenger demand.

You don't need an enormous or extremely complicated neural network.

For a college project, a well-designed forecasting model is more defensible than saying:

"We used deep learning because AI sounds cool."

A suitable model could be a regression/forecasting model, depending on the dataset you use.

6. Where does Mathematics enter?

This is where MetroMind becomes a Maths project rather than an ordinary AI project.

A. Correlation

Suppose you discover:

Rainfall $\leftrightarrow$ Passenger demand

$r = 0.72$

That suggests a strong positive relationship in your dataset.

You can also compare:

- Time $\leftrightarrow$ Demand
- Day type $\leftrightarrow$ Demand

- Temperature $\leftrightarrow$ Demand

Your syllabus specifically expects correlation and regression applications in data science and AI.

7. Regression

Regression helps model the relationship between variables and can be used for demand estimation.

For a simplified model:

$$\text{Demand} = a + b_1(\text{Time}) + b_2(\text{Rainfall}) + b_3(\text{Weekday}) + \dots$$

The system can estimate future demand from those variables.

You can show the user:

- Predicted Demand: 1,560 passengers
- Historical Average: 1,420
- Difference: +9.9%

This gives your mathematical analysis a visible purpose.

8. Probability

Prediction isn't certain.

Suppose your model predicts:

1,560 passengers

But the real number could be 1,400 or 1,750.

So the system can calculate:

What is the probability that demand will exceed available capacity?

For example:

- Bus capacity = 1,400
- Predicted demand = 1,560
- Probability of overcrowding = 81%

Your Maths syllabus includes Poisson and Normal distributions.

You could investigate whether a particular demand/arrival process is reasonably modeled using a distribution and use that model to estimate risk.

Importantly, don't force a Poisson distribution onto data without checking that it makes sense. The project should explain why a particular probabilistic model is appropriate.

9. Fourier Series

This is one of the most interesting parts.

Transport demand tends to have repeating patterns:

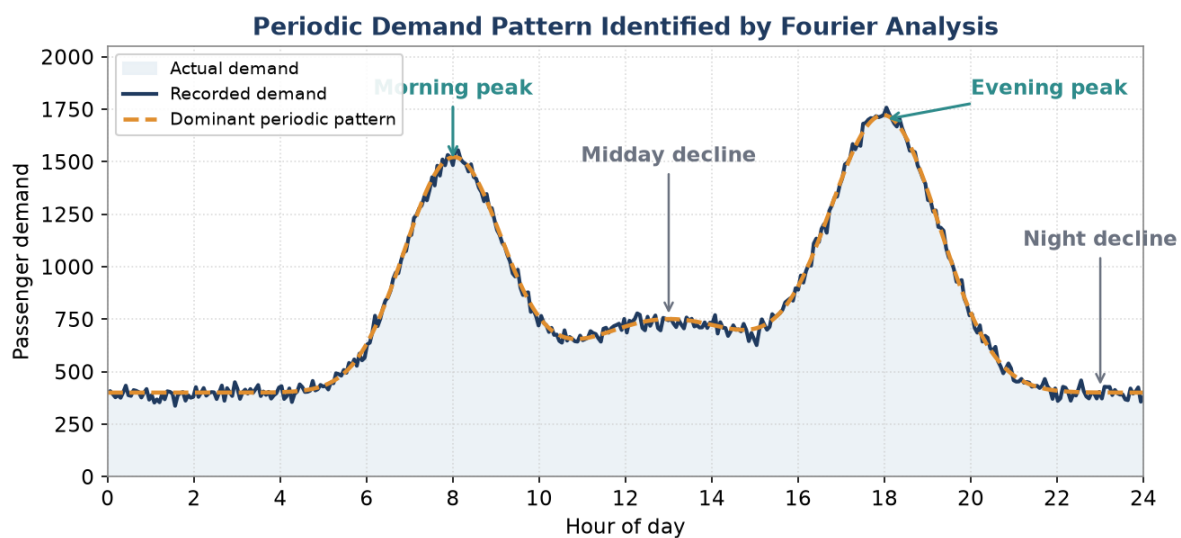
- Morning peak
- Midday decline
- Evening peak
- Night decline

And these patterns repeat across days.

Fourier analysis can help identify periodic components in a signal.

Your syllabus explicitly includes Fourier series of periodic functions.

So the app could show:



Actual demand (blue) with the dominant periodic pattern (dashed amber) detected by Fourier analysis, showing the morning and evening peaks.

You don't have to make Fourier analysis responsible for the whole forecasting system.

Instead, use it as a pattern-analysis module.

That is mathematically cleaner.

10. Optimization — the big one

Now imagine the city has:

- Available buses = 30
- Route A demand = 1800
- Route B demand = 900
- Route C demand = 600
- Route D demand = 300

You cannot send 30 buses everywhere.

The system has to answer:

How should the 30 buses be distributed?

This is an optimization problem.

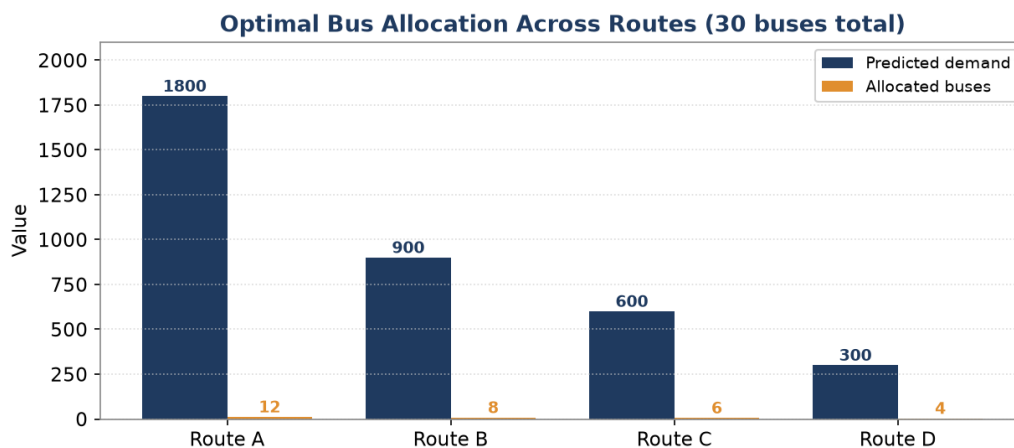
Your syllabus includes Linear Programming Problems and the Simplex method, along with nonlinear programming.

A simplified objective might be:

Minimize overcrowding + unused capacity + operating cost

subject to:

- limited buses
- route capacity
- minimum service requirements
- maximum buses per route
- total available resources



Recommended optimal allocation of 30 buses across four routes, balanced against each route's predicted demand.

The result might be:

Recommended allocation

- Route A → 12 buses
- Route B → 8 buses
- Route C → 6 buses
- Route D → 4 buses

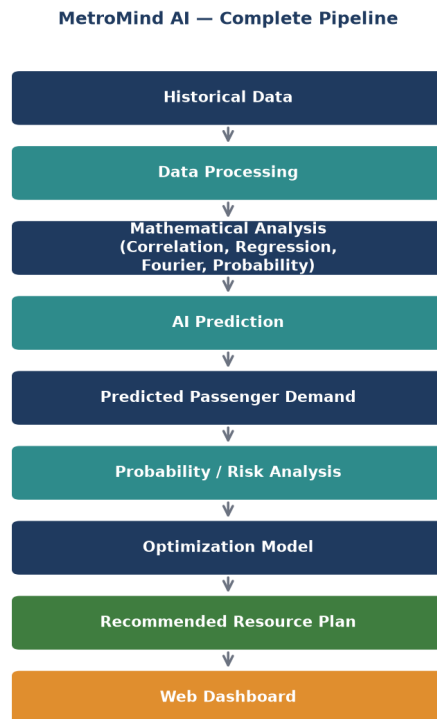
Instead of arbitrarily choosing those numbers, the system explains:

This allocation minimizes the selected objective under the defined constraints.

That is the mathematical core of the project.

11. The complete pipeline

Your architecture can be thought of like this:



End-to-end MetroMind pipeline, from historical data through mathematical analysis, AI prediction, risk analysis, and optimization to the dashboard.

This is a very strong project flow.

12. What the dashboard could look like

The dashboard should make the Mathematics visible.

Main dashboard



Mockup of the MetroMind dashboard showing predicted demand, risk summary, and a per-route recommendation table.

Then a route table:

Route	Predicted	Capacity	Risk	Recommendation
R1	1560	1400	High	+2 buses
R2	820	900	Low	Maintain
R3	1320	1100	High	+2 buses
R4	420	700	Low	Reduce

13. Add an AI Explanation panel

This is something I'd strongly recommend.

Instead of just showing:

Prediction = 1,560

show:

AI insight

"Demand on Route R1 is predicted to be 9.8% above its historical average. Monday morning and rainfall are major contributing factors in this dataset."

That makes the system feel much more intelligent.

You should still distinguish between:

- model-generated explanation
- and
- mathematical evidence actually calculated from your dataset.

14. Add a "What-If Simulator"

This could become your killer feature.

The operator can change assumptions.

Scenario 1

What if 3 buses are unavailable?

The system recalculates.

Available buses: 30 → 27

New optimal allocation:

- R1 → 11
- R2 → 7
- R3 → 6
- R4 → 3

Scenario 2

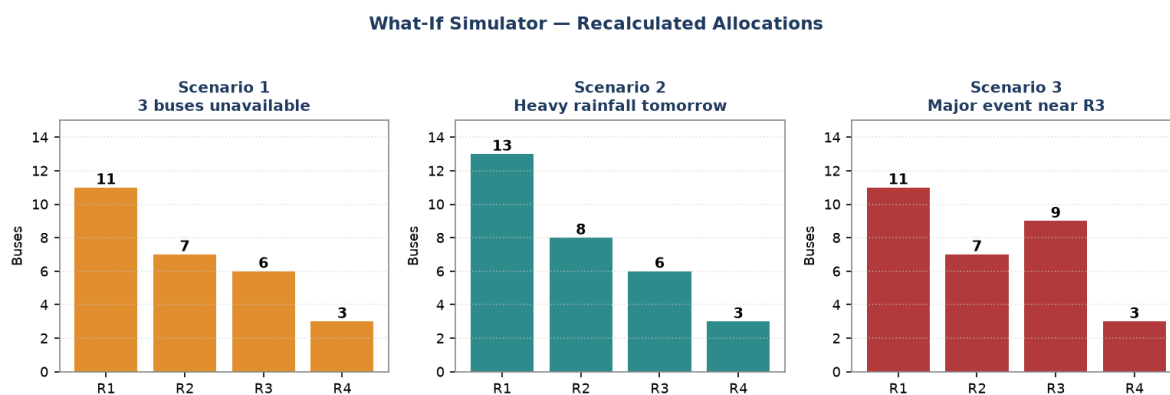
What if tomorrow has heavy rainfall?

The model adjusts predicted demand.

Scenario 3

What if a major event occurs near Route R3?

The system simulates increased demand and recalculates the allocation.



The What-If Simulator recalculates optimal bus allocation under three different scenarios.

This makes the project an actual decision-support system.

15. Add an "Algorithm/Model Comparison" page

This will be excellent for your viva.

You can compare two or more prediction models.

For example:

MODEL COMPARISON

Model	MAE	RMSE
Linear Regression	142	188
Model B	119	162
Model C	104	147

Then:

Best-performing model: Model C

This demonstrates that you aren't blindly using AI. You're evaluating it.

16. What data do you need?

You don't need a giant proprietary railway/bus dataset to create the prototype.

A dataset could contain:

- timestamp
- route_id
- station_id
- passenger_count
- day_of_week
- is_weekend
- is_holiday
- temperature
- rainfall
- special_event
- vehicle_capacity

You can build your prototype around a realistic dataset and clearly document its source.

For the academic version, data quality matters more than dataset size.

17. What you should NOT claim

This is important for the project report and viva.

Don't say:

"Our system will control Mumbai's entire public transport network."

Say:

"MetroMind AI is a mathematical and AI-based decision-support prototype for public transport demand prediction and resource allocation."

That is realistic.

Likewise, don't claim:

"AI guarantees that overcrowding will disappear."

Instead:

"The system predicts demand and recommends an optimized resource allocation intended to reduce overcrowding under defined constraints."

Much more academically sound.

18. Which Maths topics should you actually use?

You do not need to force every chapter of Maths into the project.

I'd use:

Core topics

- 1 **Correlation** — To identify relationships between demand and influencing variables.
- 2 **Regression** — For demand modeling/prediction.
- 3 **Probability** — For uncertainty and overcrowding risk.
- 4 **Fourier Series** — For periodic demand-pattern analysis.
- 5 **Linear Programming** — For optimal bus/resource allocation.

Optional advanced topic

Matrices / eigenvalues

You could represent route relationships using matrices, but I would treat this as an optional extension rather than forcing eigenvalue calculations into the main system.

Your syllabus contains matrix/eigenvalue topics, but the five areas above form a much more coherent end-to-end story.

19. Where exactly does AI fit?

This is crucial.

AI should not merely be a chatbot in the corner saying:

"Hello, how can I help?"

The AI's core job should be:

Demand forecasting

Given historical and contextual data:

$$X \rightarrow D^{\wedge}$$

where X represents the features and $D^{\wedge}$ is predicted passenger demand.

Then Maths takes that prediction and performs:

risk analysis + optimization

So:

Where AI Fits vs Where Mathematics Fits



The clear division of responsibility: AI forecasts what is likely to happen; mathematics explains the patterns, quantifies uncertainty, and optimizes resources.

That division is excellent for an AIDS + Mathematics project.

20. What technology can you build it with?

Since this is your Maths mini-project, you don't need an enormous enterprise stack.

A practical architecture could be:

Frontend

React — For dashboard and visualizations.

Backend

Python FastAPI / Flask — For AI and mathematical computation.

Data / database

SQLite or MySQL

AI / ML

Depending on the selected model:

- scikit-learn
- pandas
- NumPy

Mathematical computation

- NumPy
- SciPy
- mathematical algorithms implemented by your team

Visualization

- Plotly
- Matplotlib

You can also later integrate your Java/Spring skills, but I wouldn't make the architecture unnecessarily complicated just because your other subject has Java.

21. What would a demo look like?

This is the presentation flow I'd aim for.

- 1 **Step 1** → Upload historical transport data.
- 2 **Step 2** → System shows correlation analysis. "Rainfall and demand correlation = ..."
- 3 **Step 3** → Show periodic demand analysis. "Strong morning and evening periodic components detected."
- 4 **Step 4** → Enter: Tomorrow: Monday + Heavy Rain
- 5 **Step 5** → AI predicts: Route R1 → 1,620 passengers
- 6 **Step 6** → Probability module says: 83% overcrowding risk
- 7 **Step 7** → Optimization module calculates: Recommended allocation: +2 buses
- 8 **Step 8** → Change the scenario: 3 buses unavailable
- 9 **Step 9** → System recalculates the allocation.
- 10 **Step 10** → Show: Before optimization vs After optimization

That is the moment where the examiner understands the complete value of the project.

22. What makes it "legendary"?

Don't think of the project as:

"Passenger prediction website."

Think of it as:

A mathematical decision-support engine for public transport.

The prediction is only the beginning.

The real innovation is the chain:

Understand → Predict → Quantify uncertainty → Optimize → Recommend → Simulate

That is what elevates it.

23. Possible project title

I'd use:

MetroMind AI

AI-Driven Public Transport Demand Prediction and Mathematical Resource Optimization

Or the more academic:

MetroMind AI: An AI and Mathematical Optimization Framework for Public Transport Demand Forecasting and Resource Allocation

24. One-sentence explanation for your professor

"MetroMind AI is a decision-support application that uses AI to forecast public transport demand and mathematical techniques such as correlation, regression, probability, Fourier analysis and linear programming to identify overcrowding risk and optimize the allocation of limited transport resources."

That's the clearest description of the entire project.

25. Why this fits your Maths mini-project particularly well

Your syllabus says the course aims to use mathematics for modelling computer-engineering problems and obtaining solutions, and specifically includes optimization, statistical techniques, probability and Fourier analysis.

It also explicitly says correlation and regression can be applied to data science, machine learning and AI.

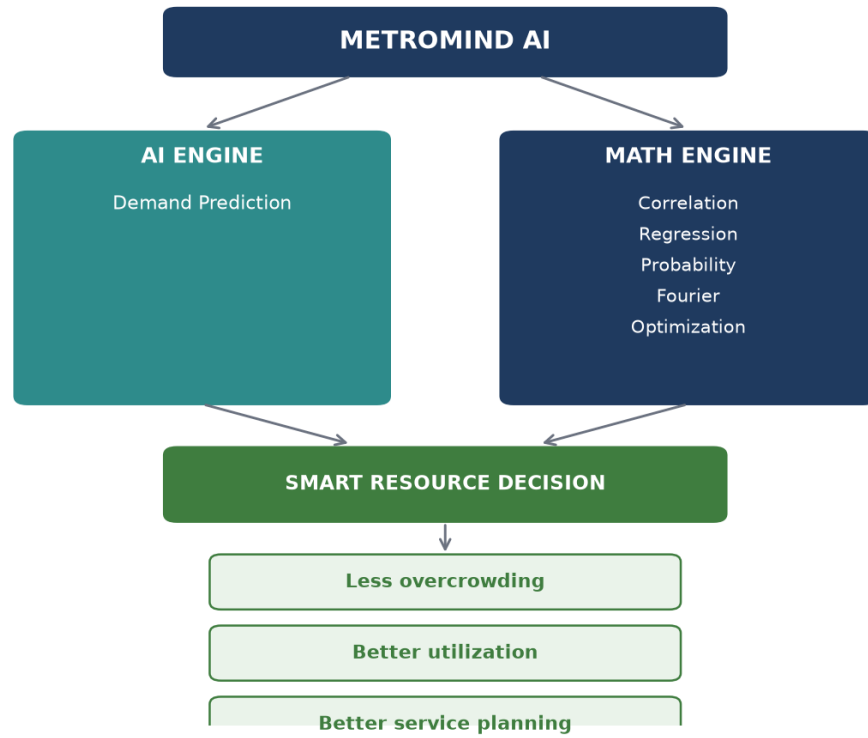
And the mini-project guidance emphasizes identifying a real need, proposing a solution, building a working model, validating it, and addressing societal problems/innovation.

So MetroMind isn't just something you can connect to the syllabus.

It naturally follows the syllabus.

The core idea in one diagram:

The Core Idea in One Diagram



MetroMind AI combines an AI engine (demand prediction) with a mathematics engine (correlation, regression, probability, Fourier, optimization) to produce smart resource decisions.

Appendix

Refining the Problem - From Bus-Level Counts to Route-Level Demand

A. But how do we know the no of passengers travelling through each bus individually?

Exactly — that's one of the biggest practical issues with the original MetroMind idea. You generally cannot know the exact number of passengers inside each individual bus just from route-level data.

But this actually gives us a chance to make the project more realistic.

We should change the problem slightly

Instead of claiming:

"MetroMind knows exactly how many people are currently inside every bus."

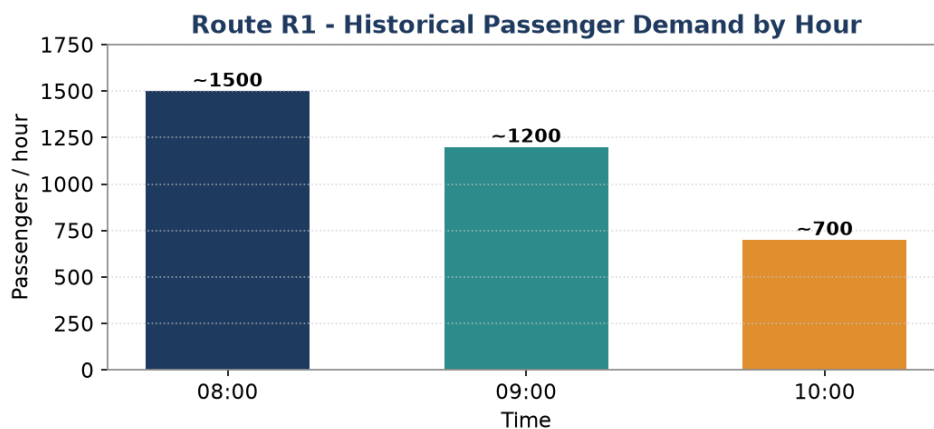
Make it:

"MetroMind predicts passenger demand at each route/stop/time and recommends how transport capacity should be allocated."

For example, historical data could tell us:

Route R1

- 08:00 → ~1,500 passengers/hour
- 09:00 → ~1,200 passengers/hour
- 10:00 → ~700 passengers/hour



Route R1 historical passenger demand by hour, as estimated from route-level data rather than individual bus counts.

MetroMind doesn't need to know that Bus #R1-023 has exactly 83 passengers.

It needs to estimate the demand that the route needs to serve.

B. But can we make it more advanced? Yes.

We could introduce passenger-count estimation.

For a prototype, imagine each bus has:

- entry/exit counter data
- ticket/boarding data
- simulated passenger-count data

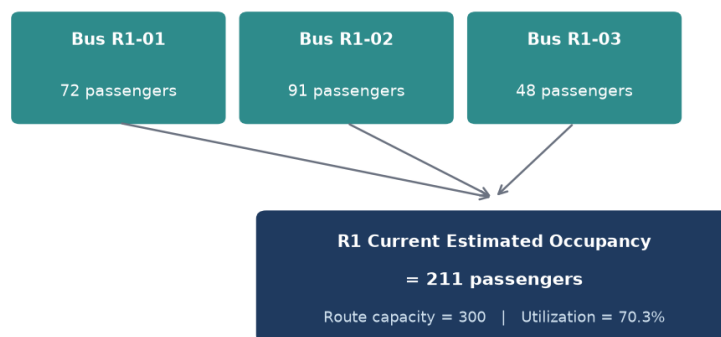
Then:

- Bus R1-01 → 72 passengers
- Bus R1-02 → 91 passengers
- Bus R1-03 → 48 passengers

The system can aggregate this:

- R1 current estimated occupancy = 211
- Route capacity = 300
- Utilization = 70.3%

Aggregating Individual Bus Counts into Route Occupancy



Individual bus counts (entry/exit or ticket data) are aggregated into a single route-level occupancy and utilization figure.

Then AI can learn from historical occupancy/demand patterns.

However, there's an important distinction

Actual passenger counting requires a data source/sensor system.

For your college prototype, you don't need to physically install sensors on buses. You can work with historical/simulated boarding and occupancy data, clearly state the assumption, and focus the project

on the mathematical prediction and optimization.

C. And I think we should make MetroMind even better

Instead of:

"How many passengers are on each bus?"

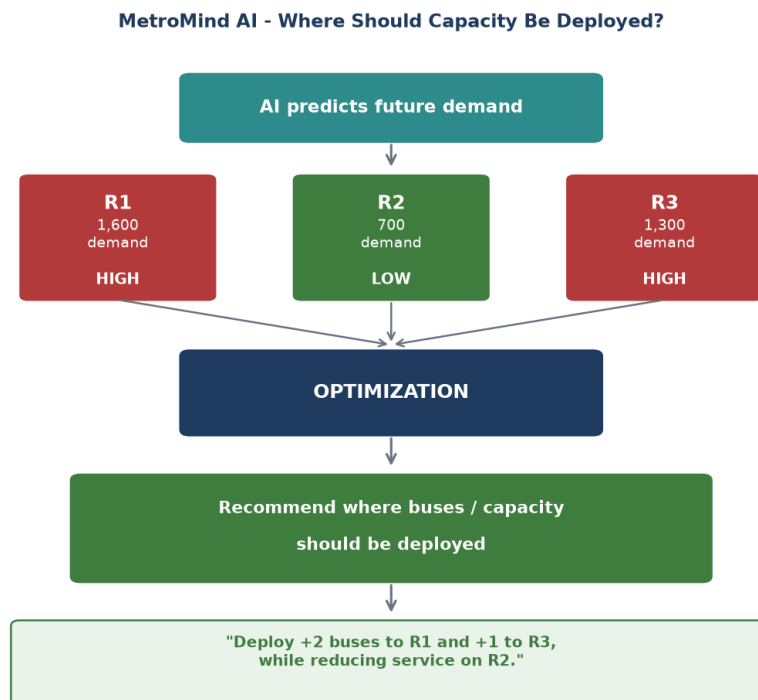
Let's make the central question:

"Given historical passenger demand, current conditions and limited buses, where should additional capacity be deployed?"

That's a much more realistic problem.

For example:

METROMIND AI



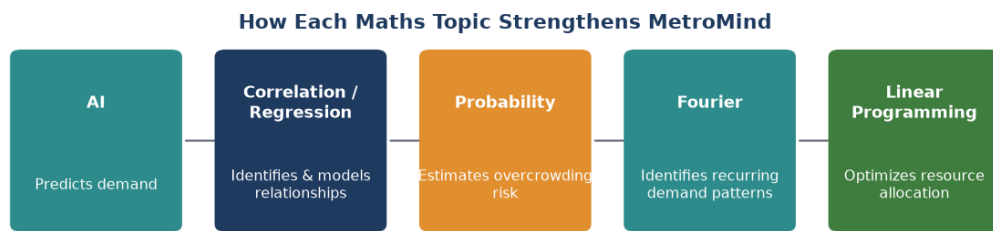
MetroMind deployment flow: AI predicts demand per route, classifies each as high or low risk, then optimization recommends where additional buses/capacity should be deployed.

So instead of pretending we have impossible real-time information, we're solving a real planning problem.

This also makes the Maths stronger:

- **AI:** predicts demand
- **Correlation/Regression:** identifies relationships and models demand
- **Probability:** estimates overcrowding risk
- **Fourier:** identifies recurring demand patterns

- **Linear Programming:** optimizes resource allocation



Each mathematical topic plays a distinct role in strengthening MetroMind, from prediction through to optimized resource allocation.

And the final output is:

"Based on the predicted demand, MetroMind recommends deploying 2 additional buses to R1 and 1 to R3 while reducing service on R2."

That's something a transport planner could actually use.

So your concern is valid — and I'd actually modify the original project specification because of it. The project should be about route/stop-level demand and capacity optimization, with individual-bus occupancy treated as optional data if available, rather than making it a required assumption.